

$i \leadsto j$ if $p_{ij}^{(n)} > 0$ for some $n \geq 0$

$i \leadsto j$ if $i \leadsto j$ & $j \leadsto i$.

Communication $\rightarrow$ equivalence relation.

Thm 1: Rec./Transience is a class property.

Pf:- If $i \leadsto j$ and i is rec, want to show j is rec.

We know $\sum_{n=0}^{\infty} p_{ii}^{(n)} = \infty$.

Enough to show $\sum_{n=0}^{\infty} p_{ij}^{(n)} = \infty$

$i \sim j \Rightarrow \exists n_1, m_1$ s.t.

$$p_{ij}^{(n_1)} > 0, \quad p_{ji}^{(m_1)} > 0.$$

$$p_{jj}^{(n+n_1+m_1)} \geq p_{ji}^{(n+m_1)} p_{ij}^{(n_1)} \geq \overline{p_{ji}^{(m_1)}} p_{ii}^{(n)} \overline{p_{ij}^{(n_1)}}$$

For any $n \geq 0$,

$$p_{ij}^{(n+n_1+m_1)} \geq p_{ii}^{(n)} \underbrace{p_{ij}^{(n_1)} p_{ji}^{(m_1)}}_{C(i,j)}$$

$$\sum_{n=0}^{\infty} p_{ij}^{(n)} = \sum_{n=n_1+m_1}^{\infty} p_{ij}^{(n)} = \sum_{n=0}^{\infty} p_{ii}^{(n)} C(i,j) \geq C(i,j) \sum_{n=0}^{\infty} p_{ii}^{(n)} = \infty$$

Periodicity of a state.

For a state i , let

$$A(i) = \{n \geq 1 : P_{ii}^{(n)} > 0\}.$$

The periodicity of a state i is defined
by the $\gcd(A(i))$. [Convention: $\gcd(\emptyset) = \infty$]
 $\lambda(i) \leftarrow$ greatest common divisor.

Consider d s.t. $d|n$ where $n \in A(i)$.
 $\sup\{d: d \text{ is a divisor of } A(i)\}$.

$$\sup(\emptyset) = \infty$$

If $n \in A(i)$, $kn \in A(i)$

$$\text{If } p_{00}^{(n)} = 0 \quad \forall n \geq 1$$

$$\Downarrow$$

$$\lambda(0) = \infty.$$

$$p_{ii}^{(n)} > 0, \quad p_{ii}^{(2n)} \geq p_{ii}^{(n)} p_{ii}^{(n)} > 0$$

$$\text{If } n, m \in A(i), \quad n+m \in A(i), \quad p_{ii}^{(n+m)} \geq p_{ii}^{(n)} p_{ii}^{(m)} > 0$$

1. RW (Simple 1dim).

$$0 \rightarrow P_{00}^{(2)} \geq P_{01} P_{10} > 0, \quad 2 \in A(0).$$

$$\lambda(0) \leq 2 \text{ as } k > 2, \quad k \not\equiv 2.$$

$$P_{00}^{(2n+1)} = 0 \Rightarrow n \in A(i) \text{ iff } n = 2m$$

Same applies for ^{higher} dim. $\Rightarrow 2 \mid n \Rightarrow \lambda(0) = 2$.
Simple RWs.

Ex Gambler's Ruin.

$$0 \rightarrow p_{00} = 1., \quad 1 \in A(0) \Rightarrow \lambda(0) = 1.$$

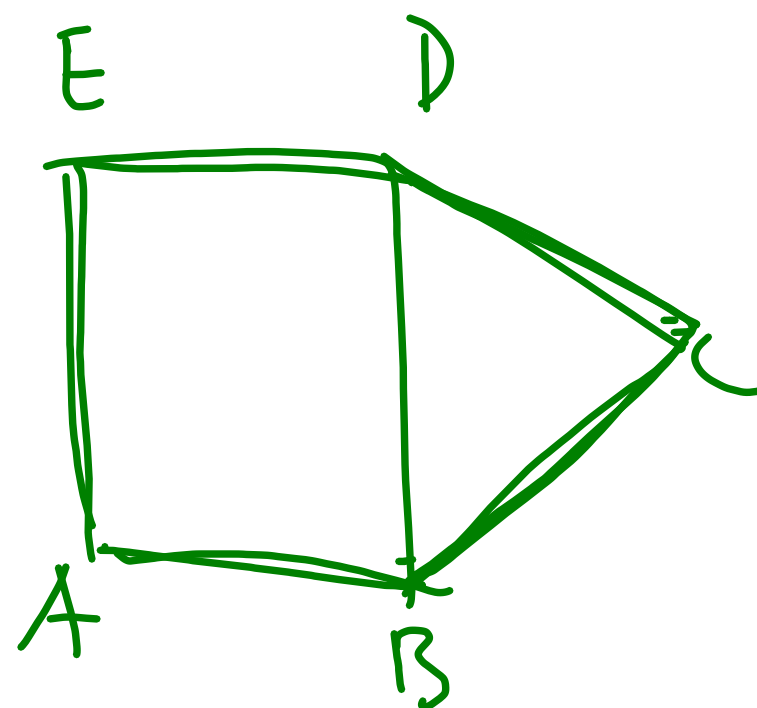
$$N \Rightarrow \lambda(N) = 1.$$

$$\text{For } 1 \leq i \leq N-1, \quad \lambda(i) = 2$$

Ex: RW on a graph.

$$p_{AA}^{(2)} \geq p_{AB} p_{BA} > 0$$

$$p_{AA}^{(5)} \geq p_{AB} p_{BC} p_{CD} p_{DE} p_{EA} > 0.$$

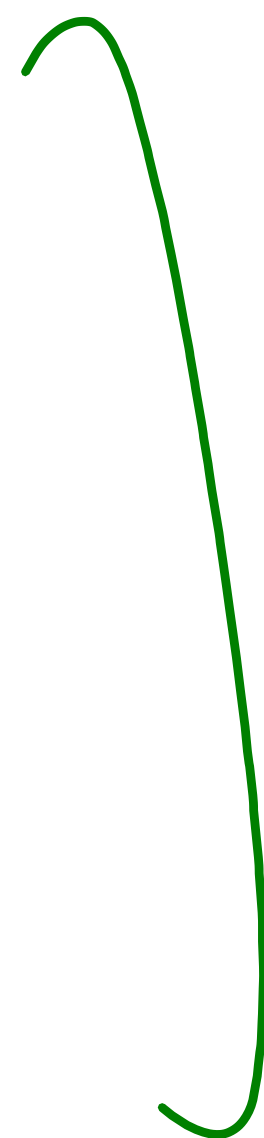


$$p_{BB}^{(2)} > 0$$
$$p_{BB}^{(3)} > 0$$

$$2, 5 \in A(A), \Rightarrow \lambda(A) = 1.$$

Reflecting RW.

$$\begin{array}{c|cccc} & 0 & 1 & 2 & \cdot & \cdot & \cdot \\ 0 & \alpha & 1-\alpha & & & & \\ 1 & q & 0 & p & & & \\ 2 & 0 & q & 0 & p & & \\ \cdot & & & & & \cdot & \cdot \\ \cdot & & & & & & \cdot \end{array}$$



→ If $\alpha = 0$,

$$\lambda(0) = 2$$

→ if $\alpha > 0$,

$$\lambda(0) = 1.$$

Thm Periodicity is also a
class property.

Pf.- Sp $i \sim j$, i.e., $p_{ij}^{(n_1)} > 0, p_{ji}^{(n_2)} > 0$.

$$A(i) = \{n \geq 1 : p_{ii}^{(n)} > 0\}, \quad A(j) = \{n \geq 1 : p_{jj}^{(n)} > 0\}.$$

Let d be a divisor of $A(i)$, if $n \in A(i)$
 $d \mid n$.

as $n_1 + n_2 \in A(i)$
 $p_{ii}^{(n_1+n_2)} \geq p_{ij}^{(n_1)} p_{ji}^{(n_2)} > 0.$

If $n \in A(i)$, then $n_1 + n_2 + n \in A(i)$.

$\boxed{p_{ij}^{(n)} \geq 0}$

$$p_{ii}^{(n_1 + n_2 + n)} \geq p_{ij}^{(n_1)} p_{ji}^{(n)} p_{ji}^{(n_2)} > 0.$$

$$\Rightarrow n_1 + n_2 + n \in A(i) \Rightarrow d \mid n_1 + n_2 + n.$$

$$\text{also } n_1 + n_2 \in A(i) \Rightarrow d \mid n_1 + n_2.$$

$$\Rightarrow d \mid n \Rightarrow d \text{ is a divisor of } A(i).$$

$$\Rightarrow \lambda(i) \text{ is a divisor of } A(i).$$

Since $\lambda(i)$ is the largest of all
divisors of $A(i)$,

$$\lambda(i) \leq \lambda(j).$$

Reversing the roles of i & j , $\lambda(j) \leq \lambda(i)$.

$$\Rightarrow \lambda(i) = \lambda(j).$$

Thm (Behaviour of $p_{ii}^{(n)}$)

a) If i is transient, then $\sum_{n=0}^{\infty} p_{ii}^{(n)} < \infty$.

b) If i rec. then $\sum_{n=0}^{\infty} p_{ii}^{(n)} = \infty$.

i) If i is null rec. then $p_{ii}^{(n)} \rightarrow 0$.

ii) If i is pos rec, with λ as the period, then

$$\lim_{n \rightarrow \infty} p_{ii}^{(n\lambda)} = \frac{\lambda}{\mu_i} > 0$$

Ex.

$$P = \begin{matrix} & \begin{matrix} 1 & 2 \end{matrix} \\ \begin{matrix} 1 \\ 2 \end{matrix} & \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \end{matrix}, \quad \begin{matrix} p_{11}^{(2n)} = 1, & p_{11}^{(2n+1)} = 0 \\ \forall n \geq 0. \end{matrix}$$

$$p_{11}^{(2n)} \rightarrow 1.$$

$$p_{11}^{(2n+1)} \rightarrow 0$$

$\lim_{n \rightarrow \infty} p_{11}^{(n)}$ does not exist.

$$\underline{\lambda(1)=2}$$

$$f_{11}^{(n)} = \begin{cases} 1 & \text{if } n=2 \\ 0 & \text{o.w.} \end{cases}$$

$$\mu_1 = \sum_{n=0}^{\infty} n f_{11}^{(n)} = 2.$$

$$= E(\tau_1 \mid X_0=1).$$

Propⁿ: Null Rec / Pos. rec is also

a class property.

Pf:- $\exists i(n)$, i.e., $p_{ij}^{(n)} > 0, p_{ji}^{(n)} > 0$.

& i is null rec. $\left(\sum_{n=0}^{\infty} p_{ii}^{(n)} = \infty, p_{ii}^{(n)} \rightarrow 0 \right)$

Want to show $\sum_{n=0}^{\infty} p_{ij}^{(n)} = \infty, p_{ij}^{(n)} \rightarrow 0$.

$$p_{ii}^{(n+n_1+m_1)} \geq p_{ij}^{(n_1)} p_{jj}^{(n)} p_{ji}^{(m_1)}$$

$$p_{jj}^{(n)} \leq \frac{p_{ii}^{(n+n_1+m_1)}}{p_{ij}^{(n_1)} p_{ji}^{(m_1)}} \rightarrow 0 \text{ as } n \rightarrow \infty.$$

$$\Rightarrow p_{jj}^{(n)} \rightarrow 0 \text{ as } n \rightarrow \infty. \quad \square$$

Propⁿ :- If $i \sim j$ and i is rec, then j must be rec.

Gambler's Ruin.

$$C_1 = \{0\}, C_2 = \{N\}, C_3 = \{1, \dots, N-1\}.$$

$$p_{10} = q > 0, \Rightarrow 1 \sim 0.$$

If '1' is rec., then $0 \sim 1 \Rightarrow 0 \prec 1$.

but $0 \in C_1, 1 \in C_3$. Contradiction.

$\Rightarrow 1$ is tran.

Ex $\hookrightarrow S = \{0, 1, \dots, 7\}$.

$C_1 = \{0, 3\}$, $C_2 = \{2, 4, 7\}$, $C_3 = \{1\}$, $C_4 = \{6\}$, $C_5 = \{5\}$

• $1 \leadsto 0 \Rightarrow 1$ is transient.

• $6 \leadsto 0 \Rightarrow 6$ is transient.

• $5 \leadsto 0 \Rightarrow 5$ is transient.

Pf of Propⁿ — $i \leadsto j \Rightarrow \exists n, \text{ s.t.}$

$$p_{ij}^{(n)} > 0.$$

i is rec.

Sp. $j \not\leadsto i$.

Therefore, for all $n \geq 0$,

$$p_{ji}^{(n)} = 0.$$

$\Rightarrow$

$$f_{ji}^* =$$

$P(X_n = i \text{ for}$

some $n \geq 1 \mid X_0 = j)$

$$\leq \sum_{n=1}^{\infty} P(X_n = i \mid X_0 = j) = 0.$$

$$i \text{ is rec.} \Rightarrow P(X_n = i \text{ infinitely many times} \mid X_0 = i) = 1.$$

$$\Rightarrow P(X_n = i \text{ for some } n > n_1 \mid X_0 = i) = 1.$$

$$= \sum_{k \in S} P(\underbrace{X_n = i \text{ for some } n > n_1}_{\text{event}}, \underbrace{X_{n_1} = k}_{\text{state}} \mid X_0 = i).$$

$$= \sum_{k \in S} P(X_n = i \text{ for some } n > n_1 \mid X_{n_1} = k, X_0 = i) \\ \times P(X_{n_1} = k \mid X_0 = i)$$

$$= \sum_{k \in S} P(\underbrace{X_n = i \text{ for some } n \geq 1}_{p_{ik}^{(n_1)}} \mid X_0 = k)$$

$$\leq \sum_{k \neq i} p_{ik}^{(n)} \times 1 + p_{ij}^{(n_1)} P(\dots)$$

$$= 1 - p_{ij}^{(n_1)} < 1. \quad \text{(contradiction)}$$